

# 軟體測試與資安檢測

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課程代號：EE 5191

授課時間：每週三 9：10—12：00

學分：3

博理館 112

民國 112 年春季班

## 期末考

考試中，請勿參閱資料。

In **Figure 1** is a procedure in C.

```
char ***alloc_array(a, b, c)
    int a, b, c;
{
    if (a <= 0 || b <= 0 || c <= 0)
        return NULL;
    r = (char ***) malloc (a * sizeof(char **));
    for (int k = 0; k <= a; k++) {
        r[k] = (char **) malloc( b * sizeof (char *));
        for (int h = 0; h <= b; h++)
            r[k][h] = (char *) malloc( (c + h + k) * sizeof(char) );
    }
    return r;
} // alloc_array( )
```

**Figure 1:** a simple C procedure

1. Draw the control-flow graph of the above program. (5pts/5)

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- This is a closed-book/closed-internet/no-sheet/no-computer exam.
- Instructor: Prof. *Farn Wang*

4. Continuing from question 2, please list all of the du-paths with respect to  $a$ ,  $b$ ,  $c$ , and  $r$ . (Note: Include all du-paths, even those that are subpaths of some other du-path). (5pts/20)
5. Continuing from question 4, please write down a minimal test set (i.e, values of  $a, b, c$ ) that satisfies all-defs coverage with respect to  $a$ ,  $b$ ,  $c$ , and  $r$ . (Direct tours only.) (5pts/25)

6. Continuing from question 4, please write down a minimal test set (i.e, values of  $a, b, c$ ) that satisfies all-uses coverage with respect to  $a$ ,  $b$ ,  $c$ , and  $r$ . (Direct tours only.) (5pts/30)

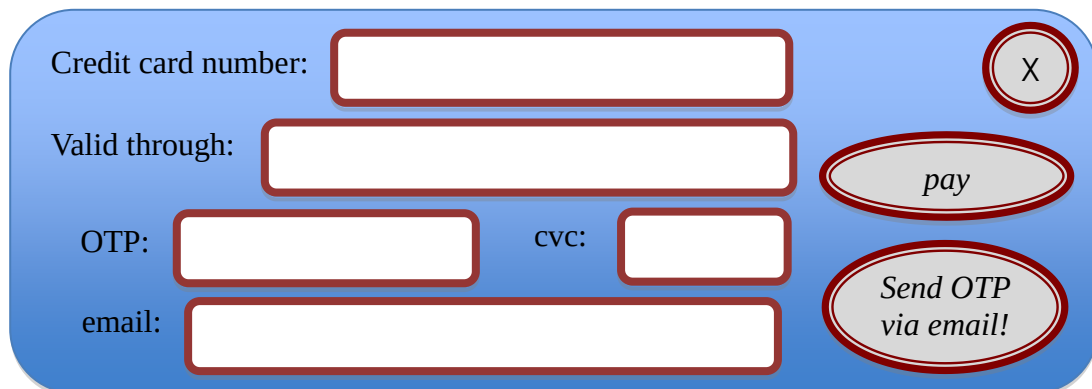
7. Consider the predicate  $\underline{a} < \underline{2} \parallel ( (!\underline{p}) \&\& (\underline{r} \parallel \underline{s} \geq \underline{t}) )$ . Please compute (and simplify) the conditions under which each of the clauses (atoms in standard logic terms) that determines this predicate. (5pts/35)

8. Continuing from question 7, write the complete truth table for all clauses (atoms in standard logic terms). (5pts/40)

9. Continuing from question 8, please identify all pairs of rows from your truth table that satisfy general active clause coverage (GACC) with respect to each clause (atoms in standard logic terms). (5pts/45)
10. Continuing from question 8, identify all rows from your table that satisfy *implicant coverage (IC)* with respect to each implicant. (5pts/50)



11. Assume that we have a web program interface for 2-step verification payment with the following screen layout.



**Figure 2:** a GUI for registration and query password.

There is no other document. Please define characteristics of inputs to the web interface and list them. (5pts/55)

12. Continuing from question 11, please present a fuzz testcase generation algorithm for the text fields according to the characteristics you defined. Please argue for your fuzzing scheme. (5pts/60)

13. In software testing, what is the difference between verification and validation?  
(3pts/63)

14. In software quality control, what is static testing? What is dynamic testing?  
(2pts/65)

15. What are the five levels of test process maturity? (5pts/70)

16. Assume that we have the procedure in Figure 1 to test. Please write down a mutant with the Arithmetic Operator Replacement (AOR) and write down a test case (values for a, b, and c) to kill the mutant. You must explain correctly why the mutant is killed by your test case. (5pts/75)
17. Assume that we have the procedure in Figure 1 to test. Please write down a mutant with the Scalar Variable Replacement (SVR) and write down a test case (values for a, b, and c) to kill the mutant. You must explain correctly why the mutant is killed by your test case. (5pts/80)

18. In Dola, Dwyer, and Soffa's 2021 paper "Distribution-Aware Testing of Neural Networks Using Generative Models", the authors explored three recent DNN testing techniques. Please explain the three techniques. (5pts/85)

19. With respect to Wu, Xu, Niu, and Nie's ICSE'22 paper "Combinatorial Testing of RESTful APIs," please explain RESTful API and constructing test transactions in a RESTful web service. (5pts/90)

20. In Ciborowska and Damevski's 2022 paper "Fast Changeset-based Bug Localization with BERT," what is a changeset? Why bug localization in changesets can be more difficult than in source code? (5pts/95)

21. In Li, Meng, Chen, Yu, Wu, Zhou, and Xu's ICSE 2022 paper "Training data debugging for the fairness of machine learning software," what is the fairness of machine learning? How should the fairness of AI models be tested? (5pts/100)